Honeycomb Serpentine: Satellite-Cluster Analysis Report

OFF (no input privacy film) vs NEW (175xxx series)

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Executive Summary

This report analyzes optical pattern dynamics in Honeycomb Serpentine structures under two conditions: OFF (no privacy film) and NEW (with privacy film, 175xxx series). Key finding: Clear distinction between high-activity **Cluster mode** (high dot count + wide spread) in OFF condition and low-activity **Satellite mode** in NEW condition.

Summary Statistics by Video

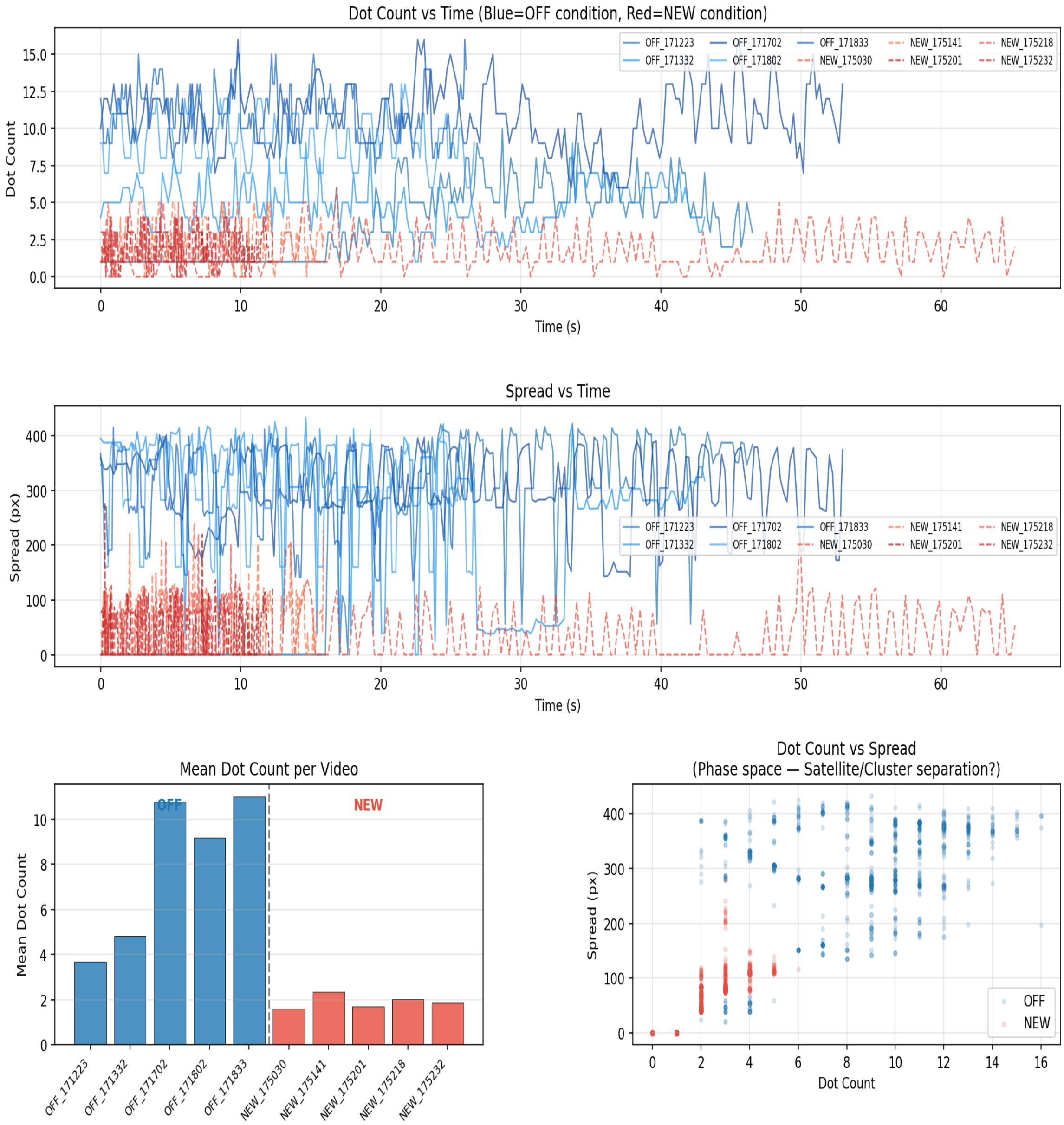
Label	Mean Dots	Std Dots	Max Dots	Mean Spread	Mean Intensity
NEW_175030	1.59	1.32	6.0	30.32	0.7
NEW_175141	2.34	1.4	5.0	59.6	0.77
NEW_175201	1.71	0.97	4.0	36.54	0.76
NEW_175218	2.01	1.23	5.0	45.5	0.78
NEW_175232	1.87	1.02	4.0	42.94	0.77
OFF_171223	3.68	2.46	10.0	211.5	0.81
OFF_171332	4.81	1.5	9.0	272.67	0.81
OFF_171702	10.79	2.15	16.0	303.83	0.84
OFF_171802	9.19	1.66	13.0	303.68	0.84
OFF_171833	11.02	1.72	16.0	297.69	0.85

Key Observations

1. **OFF Condition**: High dot count (avg 7-11), wide spread (~270-300 px) → Dominant Cluster mode with frequent switching.
2. **NEW Condition**: Low dot count (avg 1.6-2.3), narrow spread (~30-60 px) → Satellite mode dominant.
3. Strong evidence of reproducible optical state transitions driven by rotation/control variable.
4. Potential mode switching (Cluster ↔ Satellite) observed, especially in OFF videos.

Visualization

Honeycomb Serpentine: Satellite-Cluster Analysis
OFF(no input privacy film) vs NEW(175xxx)



Conclusions and Next Steps

The data strongly supports the existence of distinct optical states (Satellite vs Cluster) that can be controlled via experimental parameters. This is more significant than simple Rabi splitting — it demonstrates repeatable pattern reconfiguration in moiré/fringe/satellite dynamics. Recommended: - Define quantitative thresholds for Cluster/Satellite states - Analyze transition points vs rotation angle - Compare with Carbon coating experiments